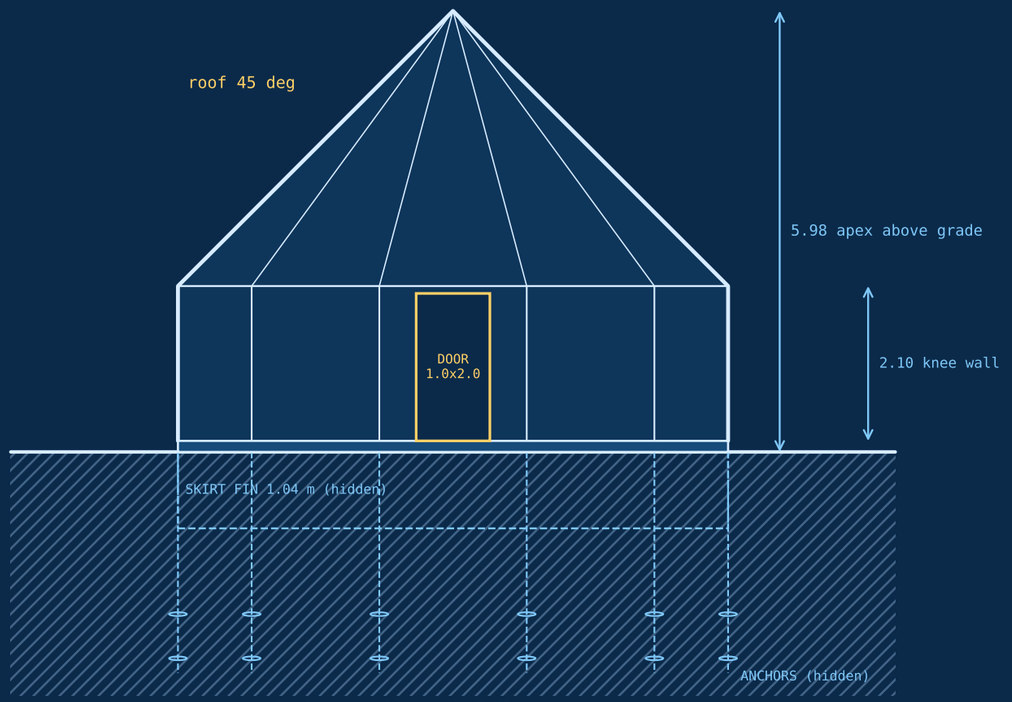
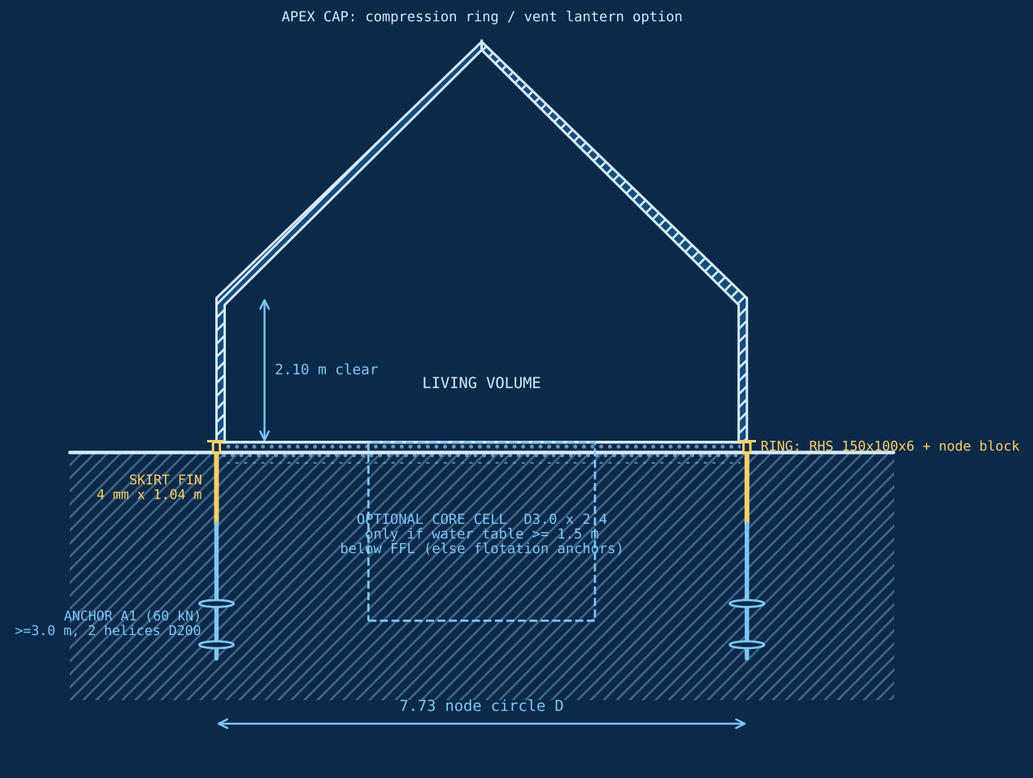
REFERENCE UNIT R-12 | D 7.73 m | 45 deg conical roof on 2.1 m knee wall | WC4 foundation

ELEVATION (view on facet S12 - door)

Facets: flat panels (12 wall + 12 roof). Hip ribs land on nodes.

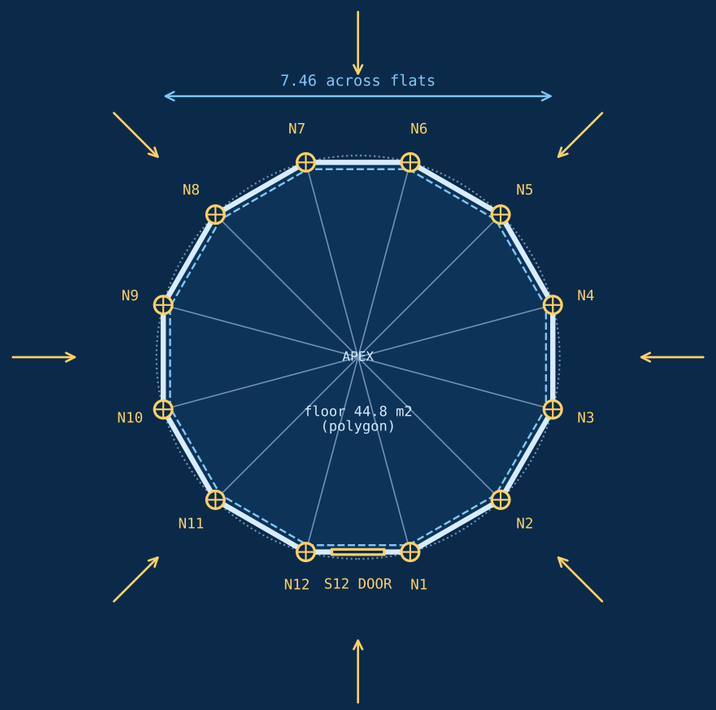


SECTION A-A (through opposite nodes N1-N7, hip line)



PLAN AT FFL (ring, nodes, anchors, hips)

WIND FROM ANY BEARING -> SAME LOAD PATH (rotational symmetry)



NOTES (concept stage; every [A] item is a verification gate - see ODF section 9)

1. Foundation = ring of 12 identical segments (chord 2.000 m) + 12 screw anchors + continuous skirt fin. Shell sits on it by 12 hip shoes.

2. Design basis WC4: 250 mph 3-s gust, ICC 500 GCpi +/-0.55 (breach case). Demand per node: T = 14.8 kN uplift; required Qult = 17.7 kN -> class A1 (60 kN).

3. Install torque target 2.6 kN.m (Qult/Kt, Kt = 23 1/m [A]); log every anchor; reject below target. Min embed >= 12 helix dia.

4. Skirt depth 1.04 m from passive resistance in dense granular soil (phi 32, gamma 18) [A]. Replace with site geotech values.

5. Loads on circumscribed circular frustum (conservative wind area). Shell areal mass 200 kg/m2 [A]. Aero Cf 0.6, CL(ext) 0.25 [A] -> wind tunnel / CFD gate V1.

6. Shelter claims require tested envelope (15 lb 2x4 @ 100 mph walls / 67 mph roof), peer review and a licensed engineer. This sheet is not that.

7. Floor at FFL = +0.15 above grade. Ring bears on compacted gravel. Berm up to knee height halves base shear (see ODF 2.6).

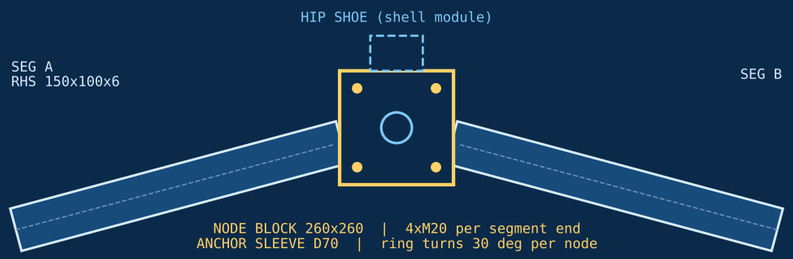
AEQUCONE v0.1 | axisymmetric shell + AEQUBASE foundation kit

SHEET 1/3 REFERENCE UNIT R-12 - ELEVATION / SECTION / PLAN

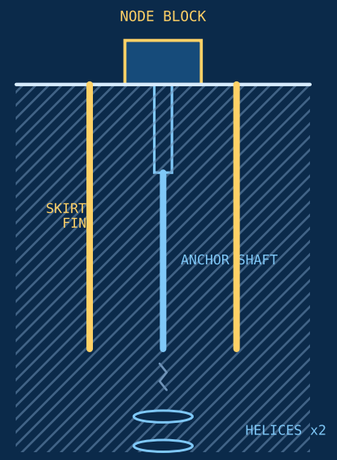
2026-10-10 | dims m | CONCEPT STAGE - SCHEMATIC - NOT FOR CONSTRUCTION

AEQUBASE FOUNDATION KIT | one segment, one node, six sizes

NODE DETAIL - PLAN (mm, schematic)



NODE SECTION



PARTS SCHEDULE - REFERENCE UNIT R-12 (same P1/P4 family for every size)

NO	PART	SPEC	QTY	kg ea	kg tot
P1	RING SEGMENT	RHS 150x100x6 x 2.000 chord + node block	12	56.8	682.1
P2	SCREW ANCHOR A1	60 kN class, >=3.0 m, 2 helices D200	12	20.0	240.0
P3	SKIRT FIN	4 mm x 2.000 x 1.04 m (trim to wind class)	12	74.9	898.5
P4	HARDWARE SET	M20 bolts, shims, torque tags	12	5.0	60.0
P5	HIP SHOE	shell interface (shell module BOM)	12	0.0	0.0
P6	ANCHOR CLASSES	A1 60 | A2 100 | A3 150 | A4 250 kN ...	0	0.0	0.0

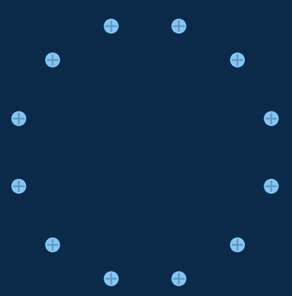
TOTAL FOUNDATION STEEL 1.88 t | gravity raft equivalent 34.3 t concrete | steel is 5.5% of raft mass

1 SET OUT



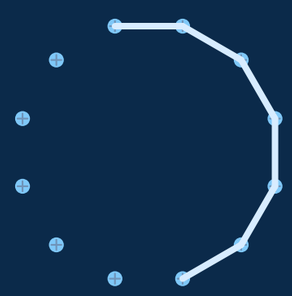
mark 12 nodes, level pad [1 h]

2 DRIVE ANCHORS



log torque >= 2.6 kN.m [4 h]

3 SET SEGMENTS



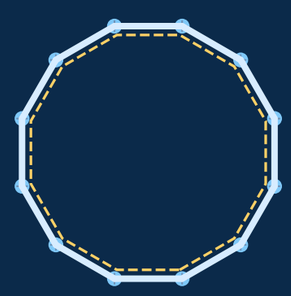
seat on sleeves [1 h]

4 CLOSE + TORQUE RING



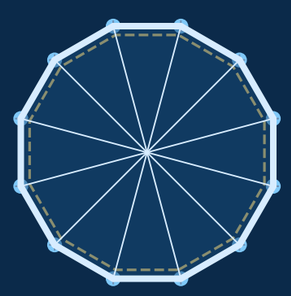
bolt up, check level [1 h]

5 FIT SKIRT + BACKFILL



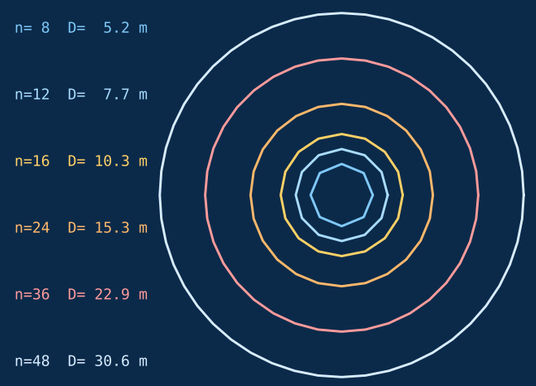
compact in lifts [1 h]

6 ERECT SHELL



hips onto shoes (shell module) [-]

SIZE LADDER - same segment, n = 8 ... 48 (nested, to scale)



FOUNDATION SELECTION CHART (WC4 = 250 mph, breach case; shell 200 kg/m2 [A])

n	D(m)	H(m)	floor	T(kN)	Qreq	anchor	skirt	steel(t)	install(h)
8	5.2	4.7	21	10.7	12.9	A1 60	0.95	1.20	6
12	7.7	6.0	47	14.8	17.7	A1 60	1.04	1.88	8
16	10.3	7.2	83	20.1	24.1	A1 60	1.12	2.60	10
24	15.3	7.5	184	31.0	37.2	A1 60	1.23	4.09	14
36	22.9	7.8	414	51.4	61.7	A2 100	1.32	6.66	20
48	30.6	8.2	734	74.7	89.6	A2 100	1.38	9.10	26

n>=24 use a truncated vented lantern (tau 0.3-0.6) to cap height.
All sizes: ONE segment section (Mseg <= 19 kN.m vs 35 kN.m design cap).

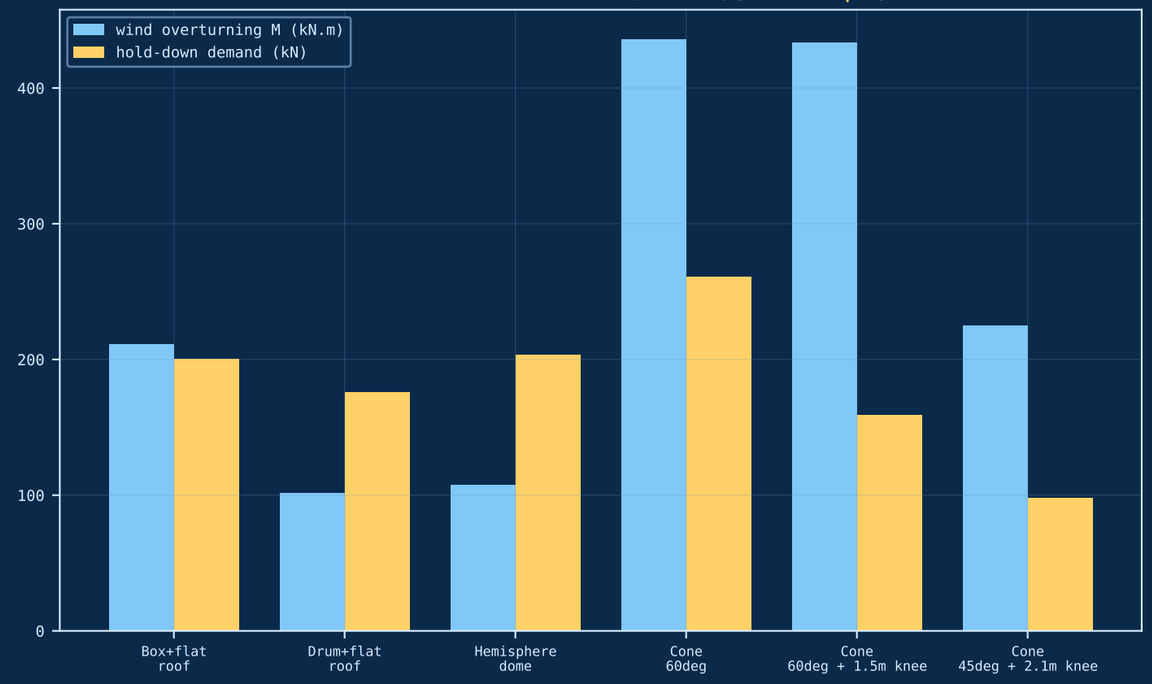
AEQUCONE v0.1 | axisymmetric shell + AEQUBASE foundation kit

SHEET 2/3 AEQUBASE KIT-OF-PARTS / ASSEMBLY / SIZES

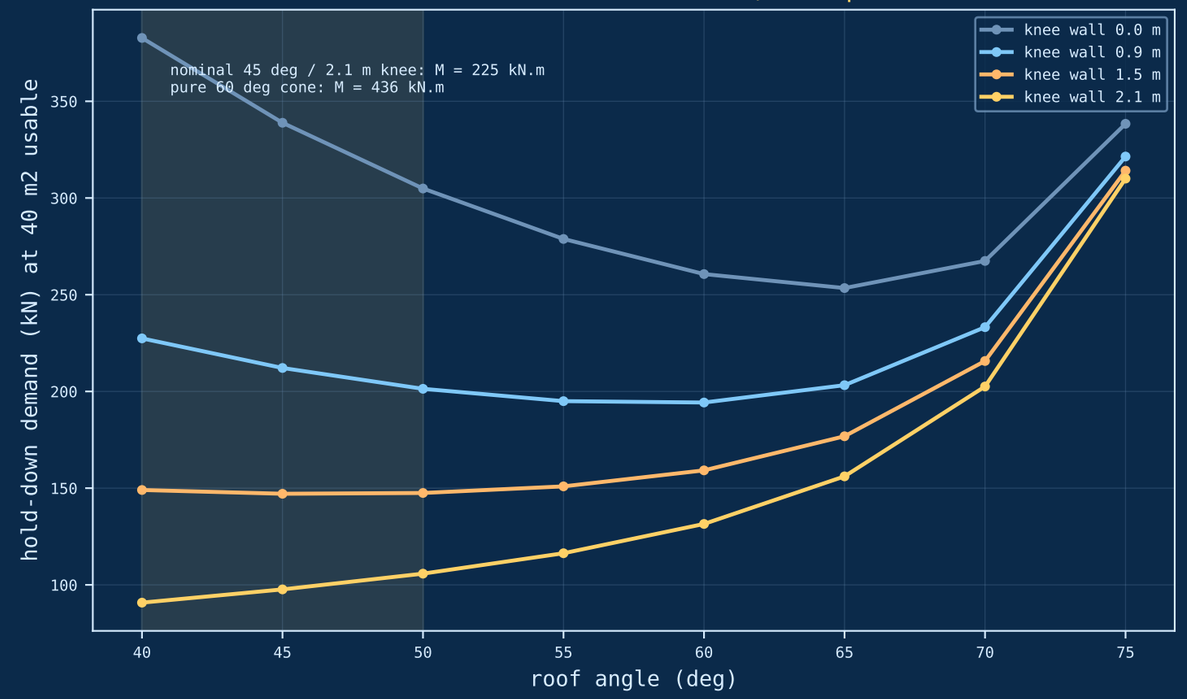
2026-10-10 | dims m | CONCEPT STAGE - SCHEMATIC - NOT FOR CONSTRUCTION

STUDY EVIDENCE | numbers behind the shape and foundation decisions (reproducible: aequcone_model.py)

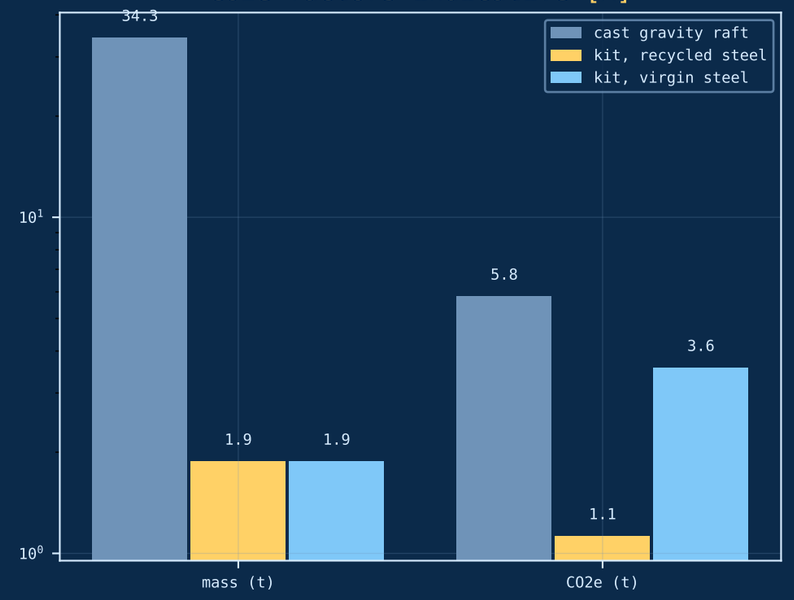
A SHAPES AT EQUAL USABLE FLOOR (40 m2), 250 mph, breach case



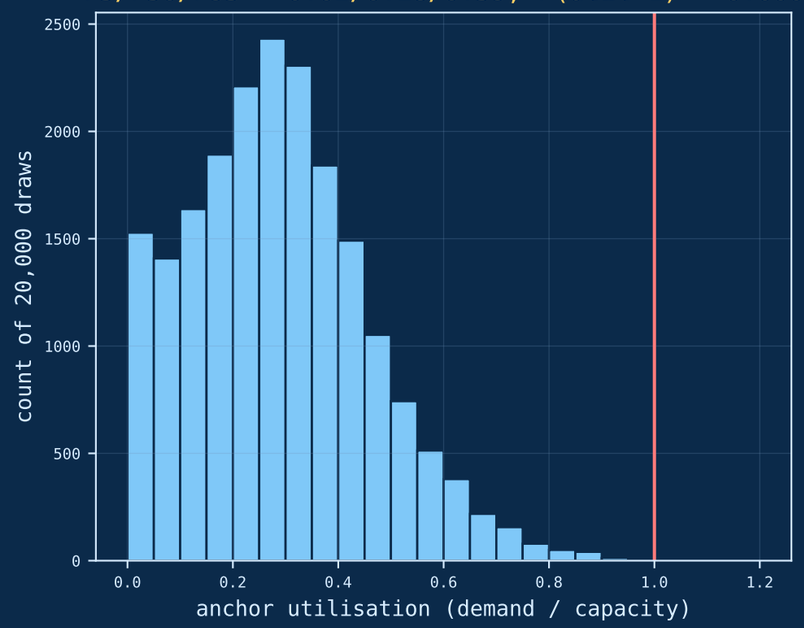
B CONE FAMILY: knee wall cuts hold-down; steep cones cost moment



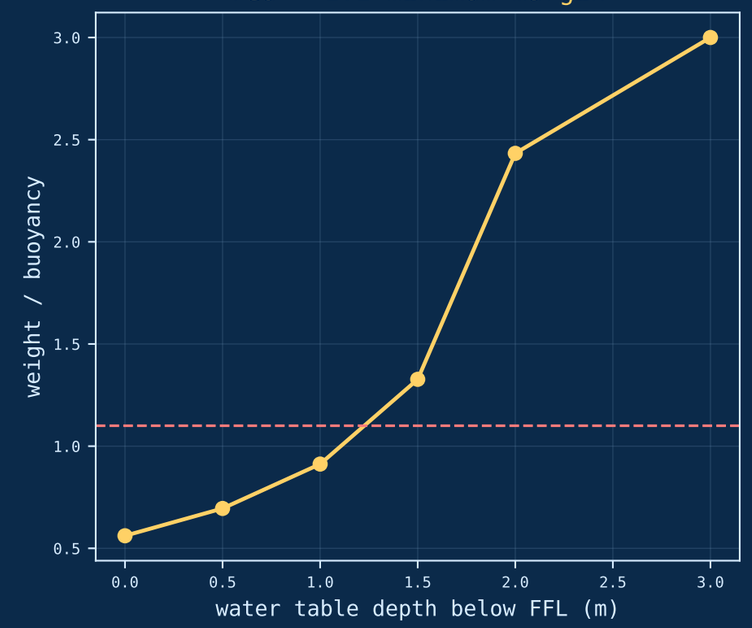
C FOUNDATION: raft vs ring + anchors (R-12, WC4)
CO2e factors indicative [A]



D MONTE CARLO (aero, soil, mass, exposure)
P5/P50/P95 = 0.02/0.28/0.59; P(util>1) = 0.000



E SUB-GRADE CORE CELL flotation (D3.0 x 2.4)
red = 1.1 minimum margin



AEQUCONE v0.1 | axisymmetric shell + AEQUBASE foundation kit

SHEET 3/3 STUDY EVIDENCE

2026-10-10 | dims m | CONCEPT STAGE - SCHEMATIC - NOT FOR CONSTRUCTION